ECON1061

Forecasting and Quantitative Analysis

Assessment 3 Semester 1, 2026

Monthly Short-Term Visitor Arrivals from South Korea to Australia

January 1991 to December 2019

Name Antony Wibgha **Student ID** s4233062

PART A - DATA DESCRIPTION, TRANSFORMATION AND BENCHMARK MODEL

Question 1

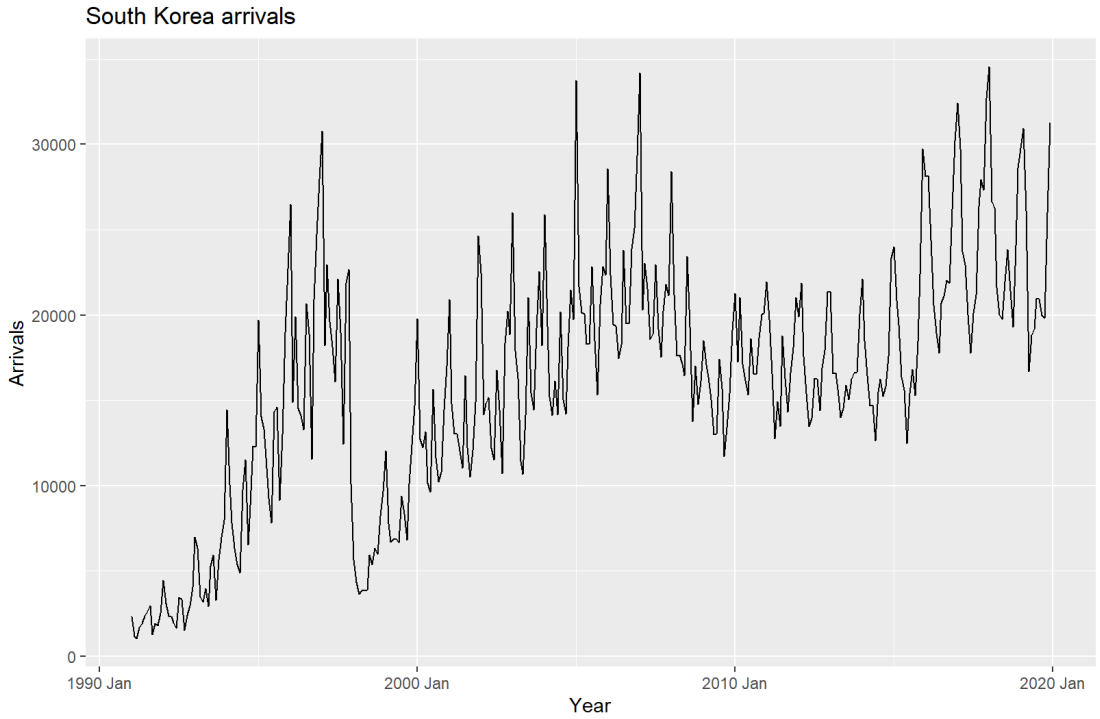


Figure 1. Time plot of monthly short-term visitor arrivals from South Korea to Australia, January 1991 to December 2019.

The series covers nearly three decades of data. Korean outbound tourism expanded sharply after 1988, when the government lifted long-standing restrictions on overseas travel following the Seoul Olympics, and arrivals to Australia grew in step with that liberalisation. Two events interrupt the upward path. First, a steep contraction from 1997 to 1999 reflects the Asian Financial Crisis: the Korean won fell roughly 50% against the Australian dollar, making long-haul travel unaffordable for most Korean households. Second, a shallower dip around 2002 to 2003 corresponds to the SARS outbreak. The seasonal swings also grow wider as arrivals increase, pointing to non-constant variance and the need for a stabilising transformation before modelling.

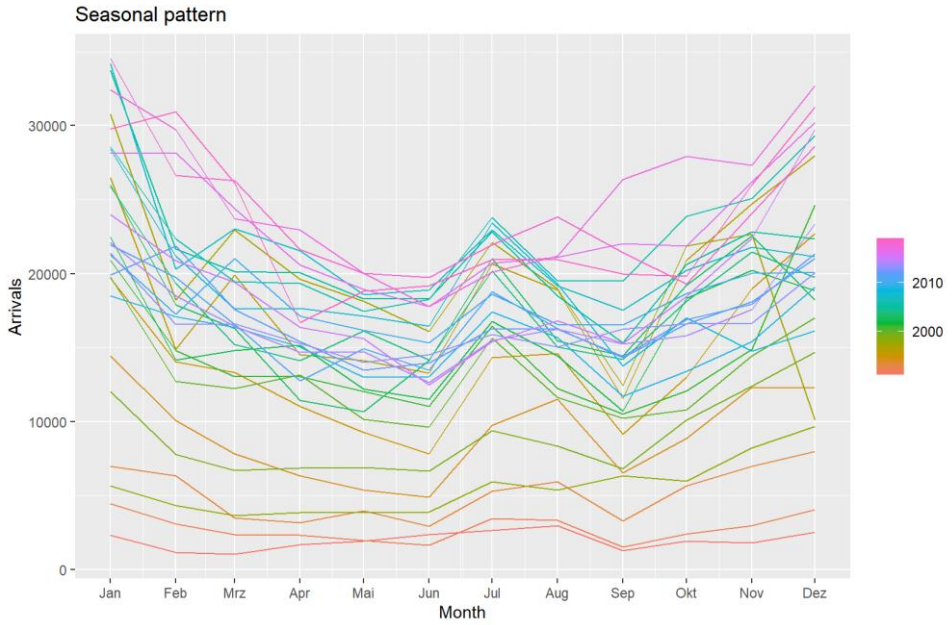


Figure 2. Seasonal plot of South Korean visitor arrivals by month.

Each year traces its own line across the twelve calendar months. Arrivals peak in January and rise again in December, fall through April and May, then show a moderate secondary lift in July before recovering to the year-end high. Later years sit above earlier ones at every month, confirming the seasonal cycle operates around a rising trend. The W-shaped profile is consistent with Korean university and school holiday calendars aligning with Australian summer, with the mid-year lift in July matching the Korean summer break.

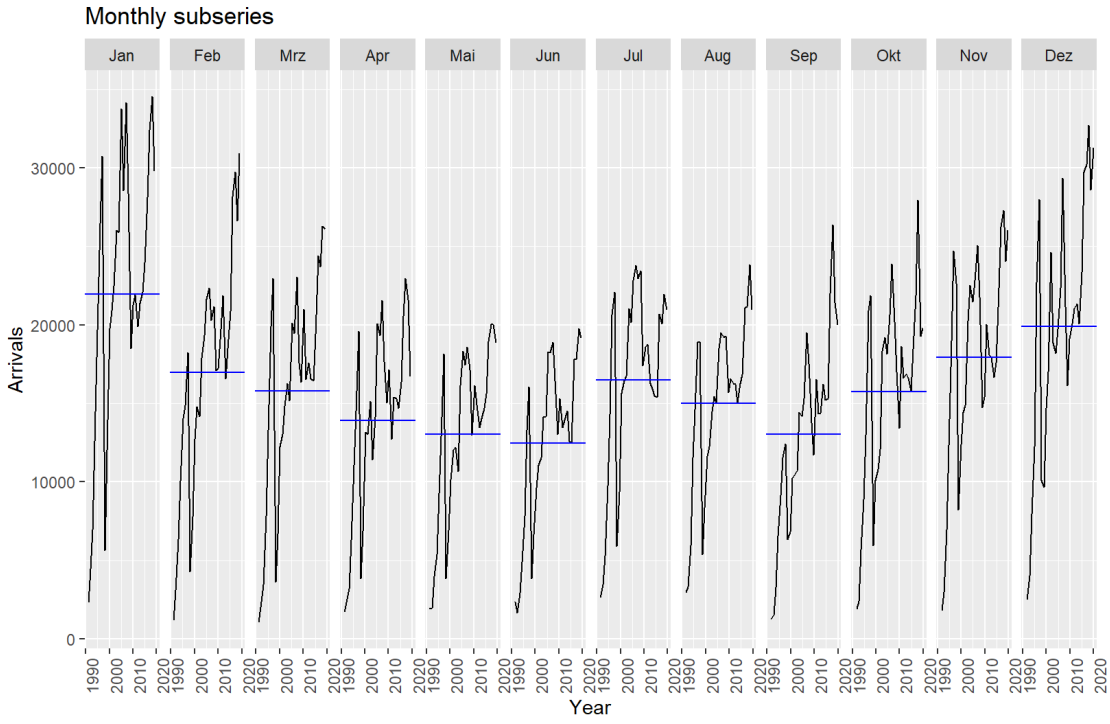


Figure 3. Subseries plot of South Korean visitor arrivals by calendar month.

Each panel shows one calendar month from 1991 to 2019. The upward slope within every panel confirms the trend runs through all twelve months. The horizontal blue line, marking the

monthly mean, is highest for January and December and lowest for May and June, consistent with Figure 2. The sharp downward steps around 1997 to 1999 correspond to the Asian Financial Crisis, and the gradual recovery that follows is visible in every month.

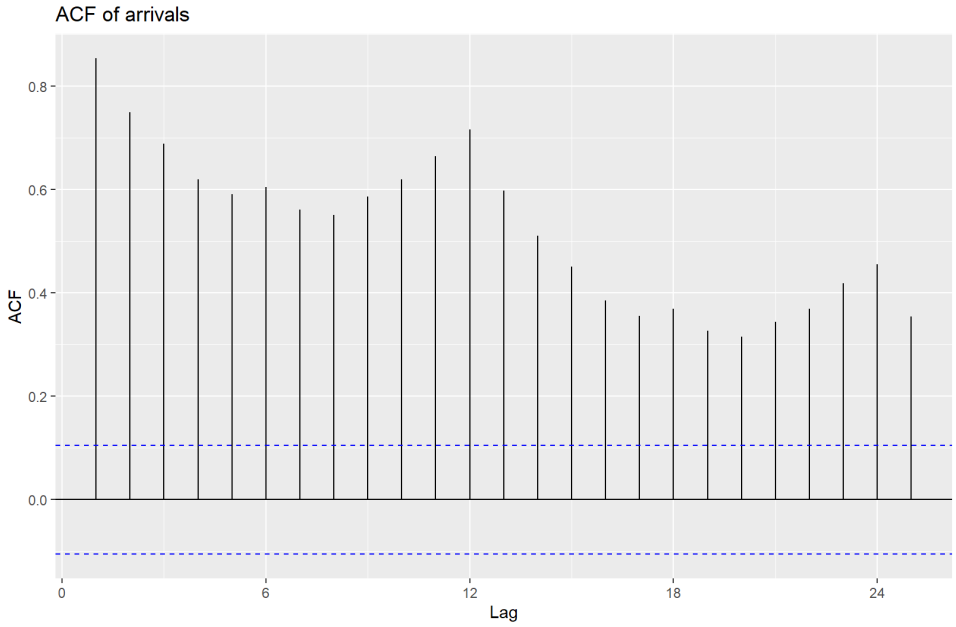


Figure 4. Autocorrelation function of monthly South Korean visitor arrivals.

Every bar exceeds the 95% significance boundary, and values decay slowly from around 0.85 at lag 1 to around 0.35 at lag 24. That gradual, hyperbolic decay is the fingerprint of a non-stationary trending process. Local peaks at lags 12 and 13, with the lag 12 bar elevated to approximately 0.71 above its neighbours, provide formal confirmation of the annual seasonal dependence seen in Figures 2 and 3. Both the trend and the seasonality must be addressed before fitting any ARIMA model.

Question 2

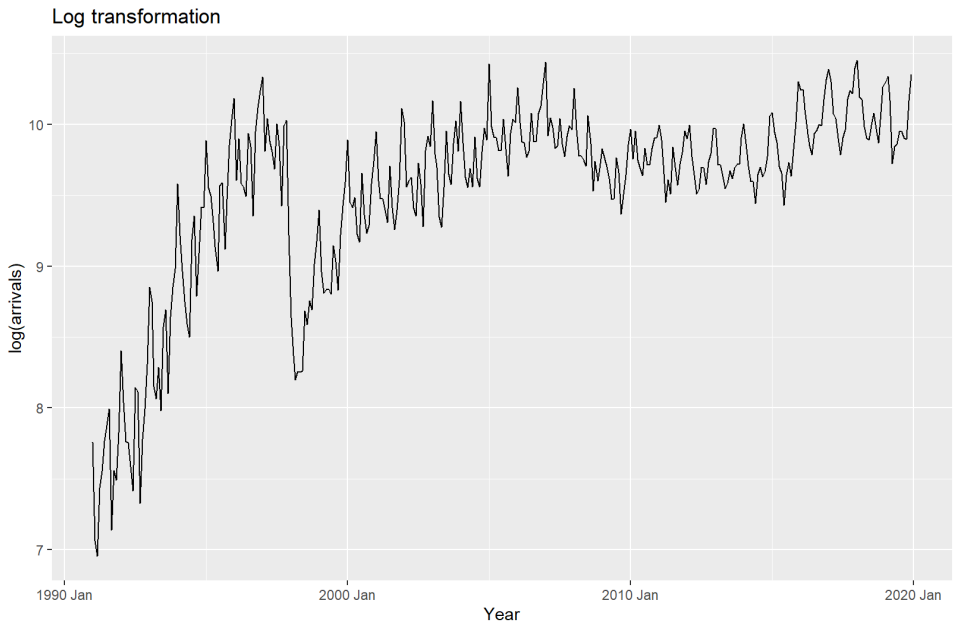


Figure 5. Log-transformed South Korean visitor arrivals.

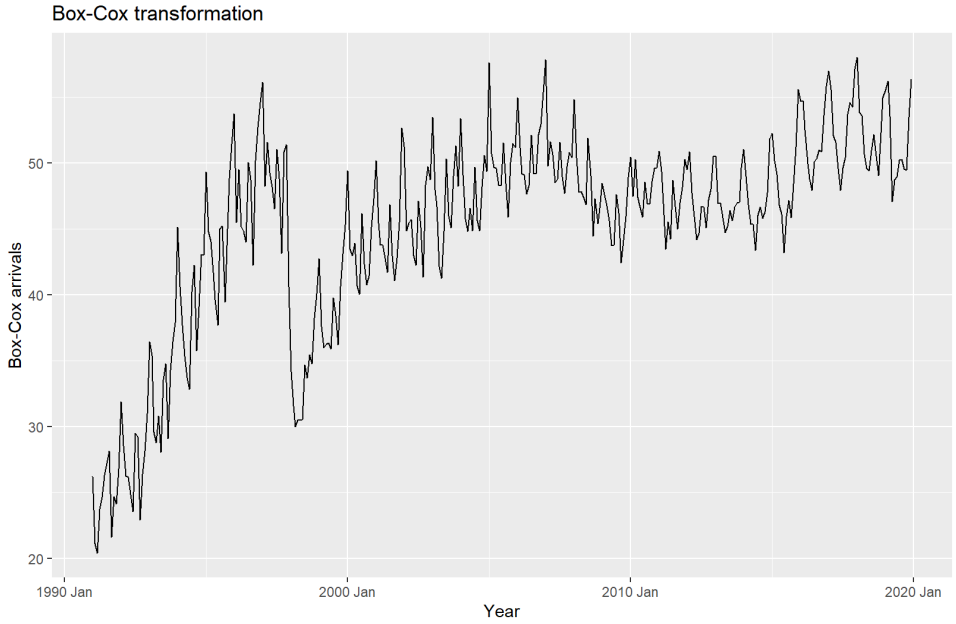


Figure 6. Box-Cox transformed South Korean visitor arrivals (Guerrero lambda = 0.2688).

The Guerrero method selected lambda = 0.2688, meaningfully above zero. Because the log transformation assumes lambda = 0, it over-compresses the early low-count portion of the series. In Figure 5, the within-year seasonal amplitude during 1991 to 1996 appears slightly narrower relative to the post-2000 period, indicating residual variance instability. In Figure 6, the seasonal amplitude in the early 1990s is proportionally similar to the rest of the sample, confirming better variance stabilisation. The drop associated with the 1997 to 1999 Asian Financial Crisis is also more symmetrically represented on the Box-Cox scale than on the log scale. The Box-Cox transformation with lambda = 0.2688 is therefore applied at every subsequent modelling stage.

Question 3

A seasonal naive model with drift is the benchmark. The time plot (Figure 1) documents persistent growth over the full sample, so the drift term is needed: without it, the model would project a flat seasonal repeat inconsistent with the observed trend. The fitted drift is 0.9655 (standard error 0.2413) on the Box-Cox transformed scale. The seasonal and subseries plots (Figures 2 and 3) confirm a stable, repeating monthly pattern, making the seasonal naive the appropriate baseline: it forecasts each month using the observed value from the equivalent month one year earlier. The Box-Cox transformation with lambda = 0.2688 is applied throughout.

Question 4

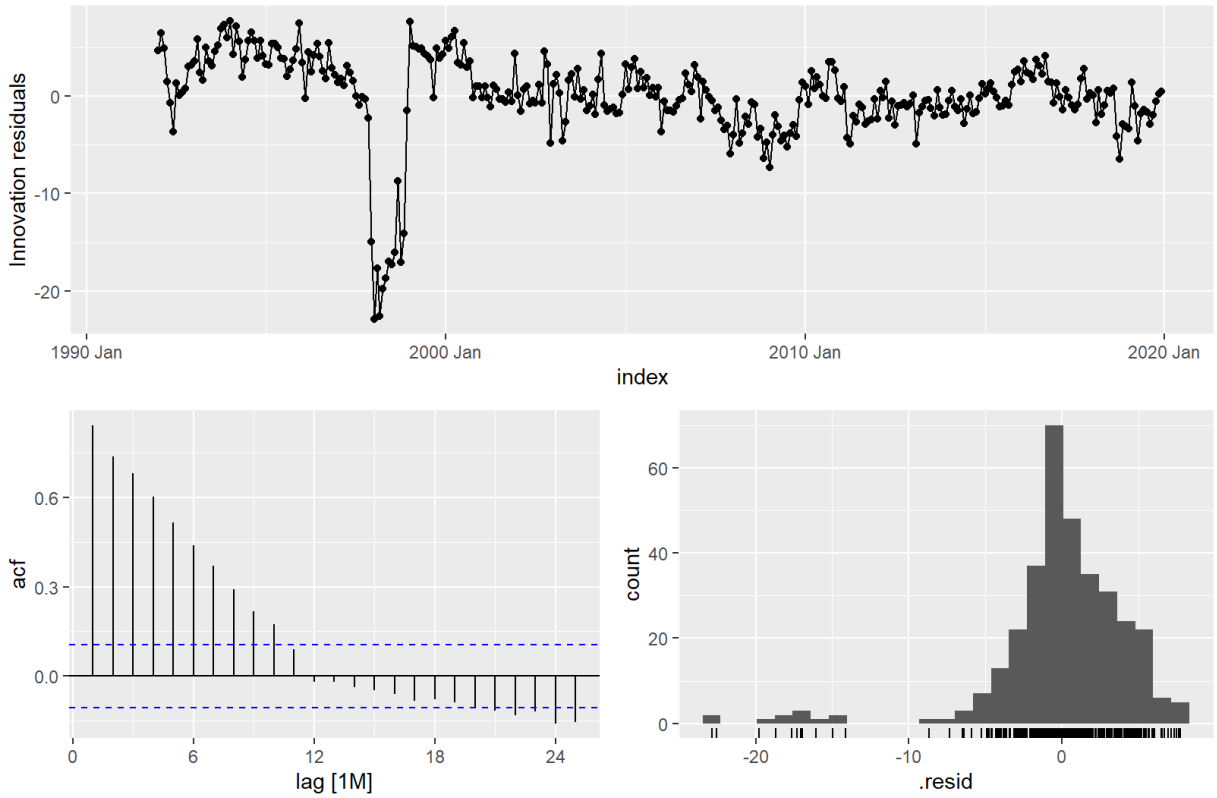


Figure 7. Residual diagnostics for the seasonal naive model with drift.

The residual time plot shows the series sitting above zero for the first part of the sample, then dropping sharply to around -25 on the transformed scale during 1997 to 1998 before recovering. The positive bias in the pre-crisis period means the model systematically under-forecast arrivals during the rapid growth phase. The ACF panel confirms strong leftover autocorrelation: bars from lag 1 through lag 12 and beyond all exceed the significance bounds, with slow hyperbolic decay starting from around 0.66 at lag 1. The histogram is skewed left, with a heavy tail extending below -20 from the crisis residuals. The Ljung-Box statistic is 973.0 with p-value less than 0.001 (dof = 6, lag = 12), so the white noise null is strongly rejected. These properties are expected for a benchmark model applied to a series with strong trend and seasonal structure.

Question 5

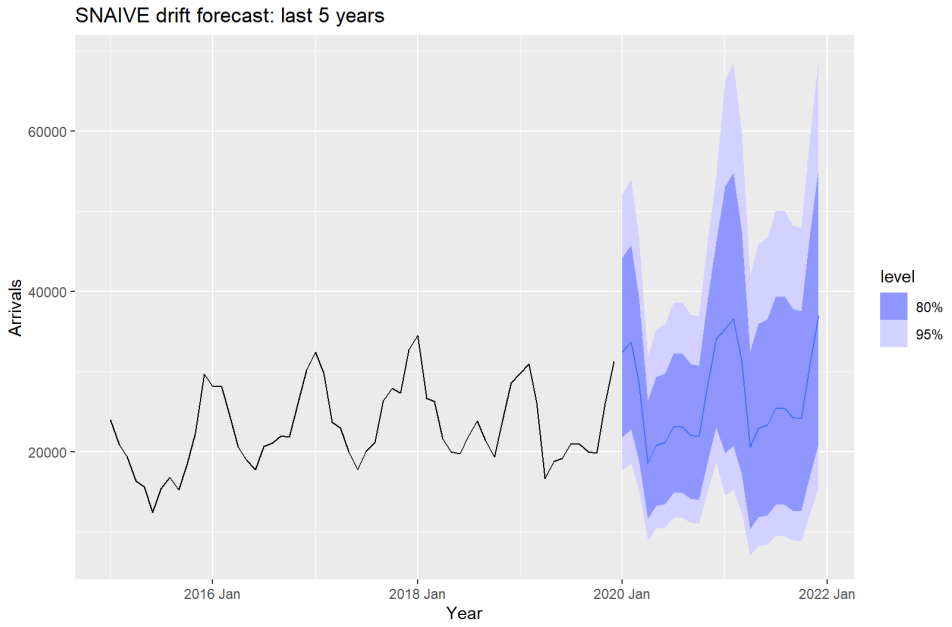


Figure 8. Two-year forecasts from the seasonal naive with drift model, last five years of history shown.

The model projects the 2019 seasonal profile forward one year and repeats it in 2021, with a small upward drift added each period. The W-shaped seasonal cycle is preserved because the model anchors directly on the previous year's observed values. Prediction intervals widen progressively, with the 95% band noticeably wider than the 80% band by the end of the second year. The forecast horizon covers 2020 to 2021, coinciding with the COVID-19 pandemic, which caused a near-complete halt in international arrivals. No univariate model trained on pre-2020 data could anticipate that outcome, illustrating the fundamental limits of mechanical extrapolation when structural breaks occur.

PART B - ARIMA AND ETS MODELS

Question 6a

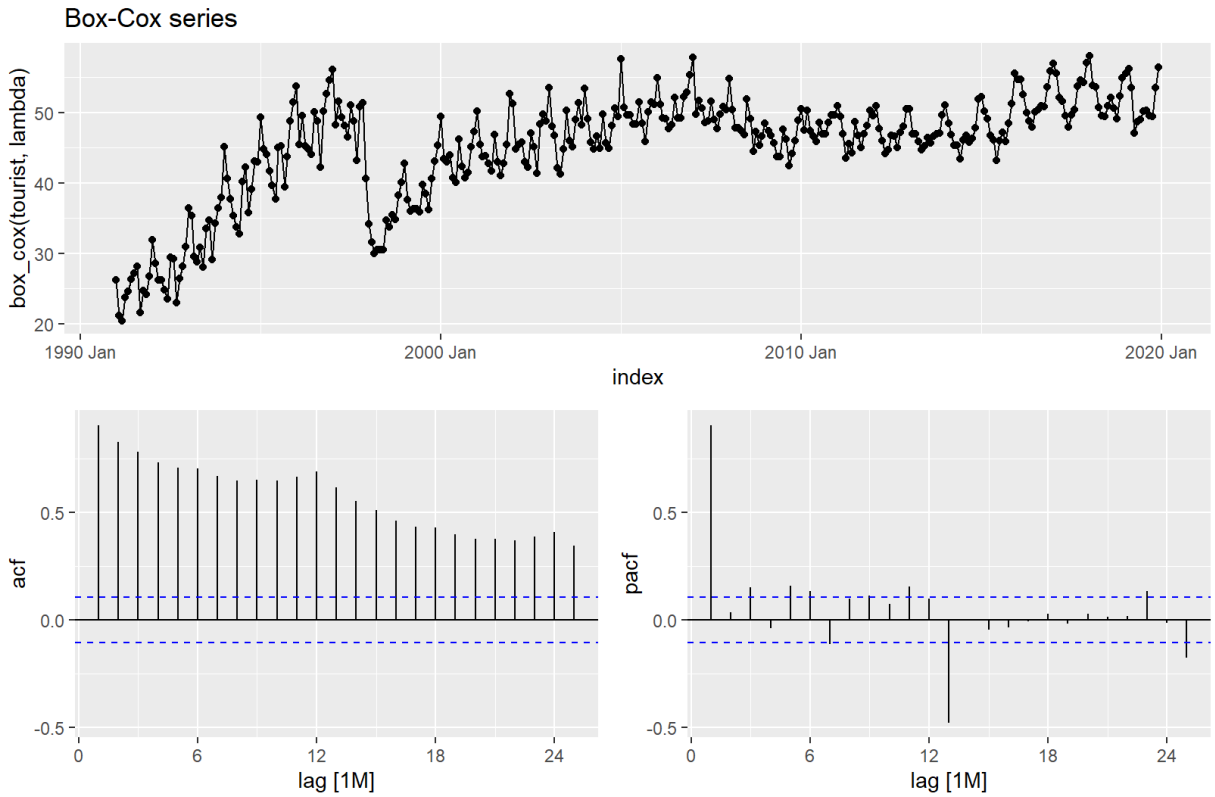


Figure 9. Time plot, ACF and PACF of the undifferenced Box-Cox transformed series (Step 1).

The undifferenced series displays all three signs of non-stationarity. The time plot trends upward throughout. The ACF decays very slowly from around 0.80 at lag 1, with every bar above the significance bounds and local peaks at lags 12 and 24. The PACF has a single dominant spike at lag 1 near 0.80 followed by rapid decay, the pattern expected for a near-unit-root process. The KPSS statistic is 3.16 with p-value at or below 0.01, rejecting stationarity. Automatic unit-root tests confirm ndiffs = 1 and nsdiffs = 1, so both one non-seasonal and one seasonal difference are required.

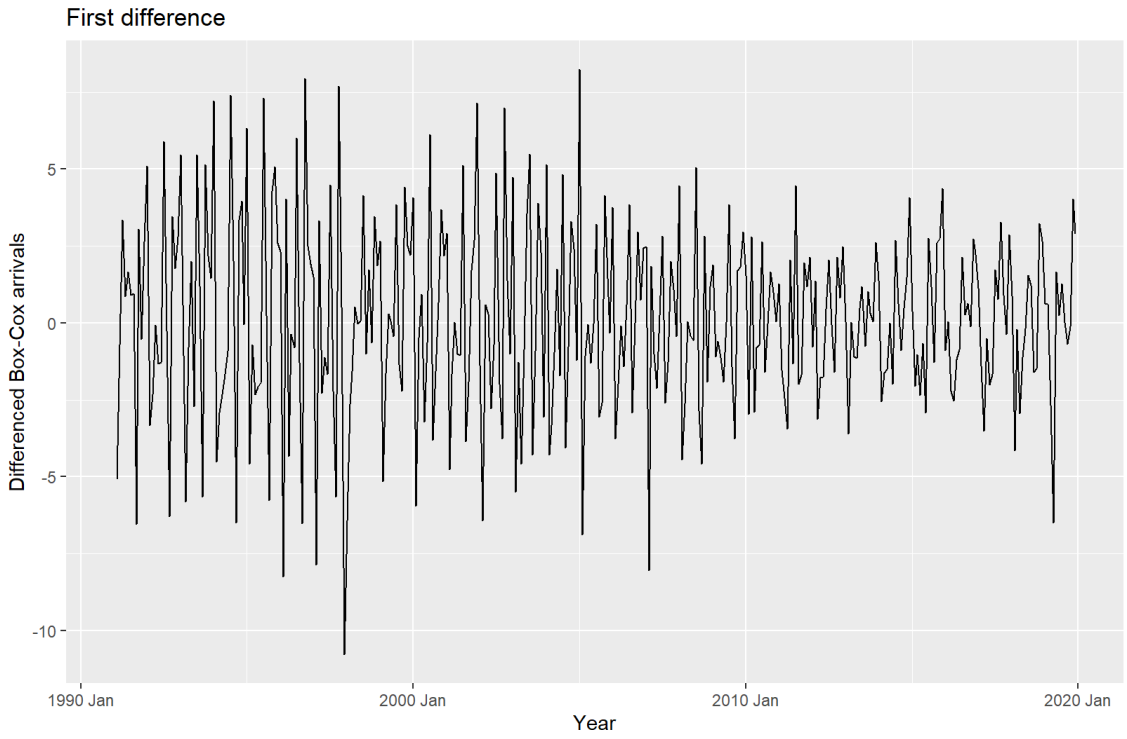


Figure 10. First-differenced Box-Cox transformed series (Step 2).

One first difference removes the long-run upward trend: the series now oscillates around zero with no directional movement. A strong seasonal wave with a 12-month period remains clearly visible, with large amplitude relative to the mean. First differencing alone is insufficient for this monthly arrivals series.

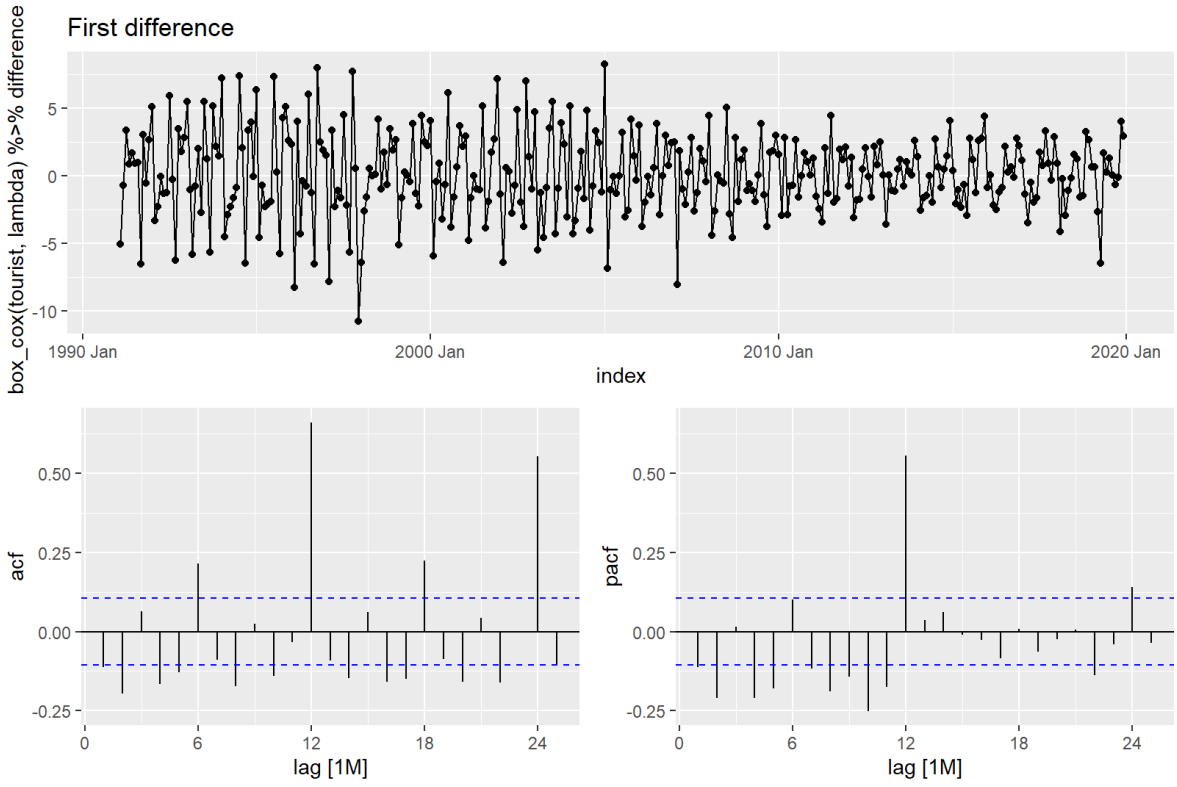


Figure 11. ACF and PACF of the first-differenced series (Step 2).

After first differencing, the ACF retains large spikes at lags 12 and 24 of approximately 0.55, both well above the significance bounds. The PACF has a corresponding spike at lag 12 near 0.55. This seasonal autocorrelation structure confirms the annual cycle has not been removed by first differencing alone. A seasonal difference is also required.

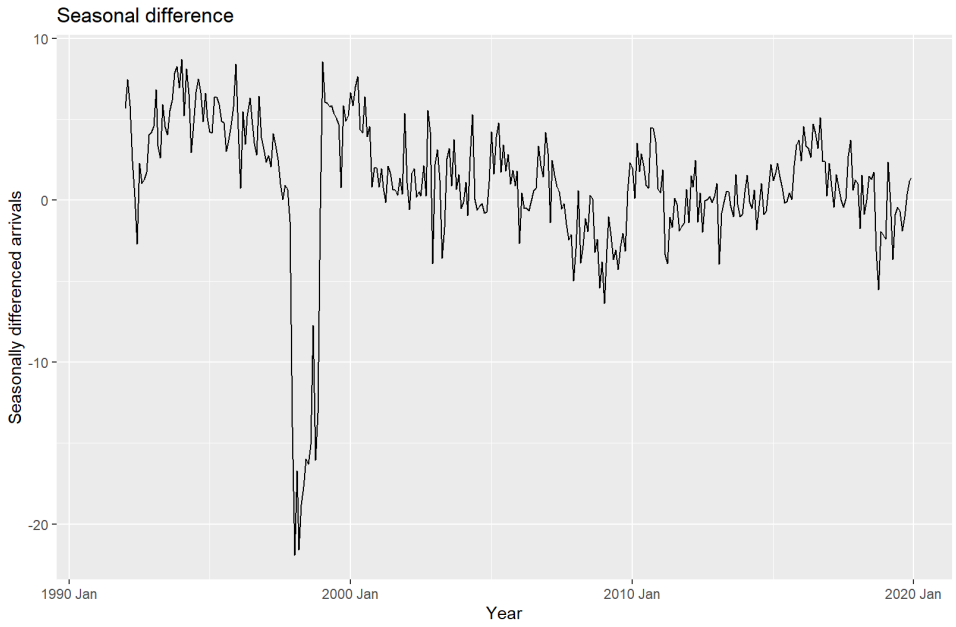


Figure 12. Seasonally differenced Box-Cox transformed series (Step 3).

One seasonal difference at lag 12 removes most of the annual repetition and the within-year wave largely disappears. The series sits closer to zero on average, but a positive drift remains visible in the early part of the sample and a large shock appears during 1997 to 1998. These features indicate non-seasonal trend persistence after seasonal differencing alone.

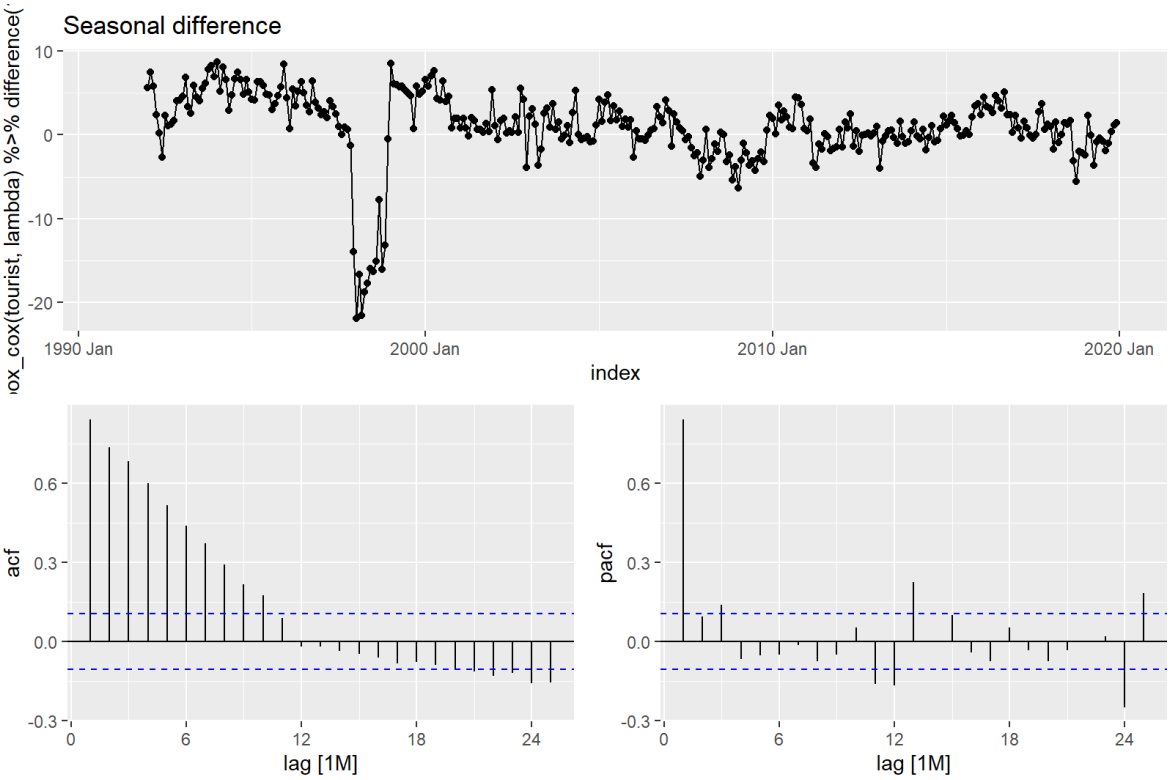


Figure 13. ACF and PACF of the seasonally differenced series (Step 3).

After seasonal differencing the ACF decays slowly from around 0.67 at lag 1, with bars remaining above the bounds for the first several lags. The PACF has a dominant spike at lag 1 near 0.67. This mirrors the undifferenced pattern in Figure 9 and confirms that non-seasonal non-stationarity persists. A first difference is also needed.

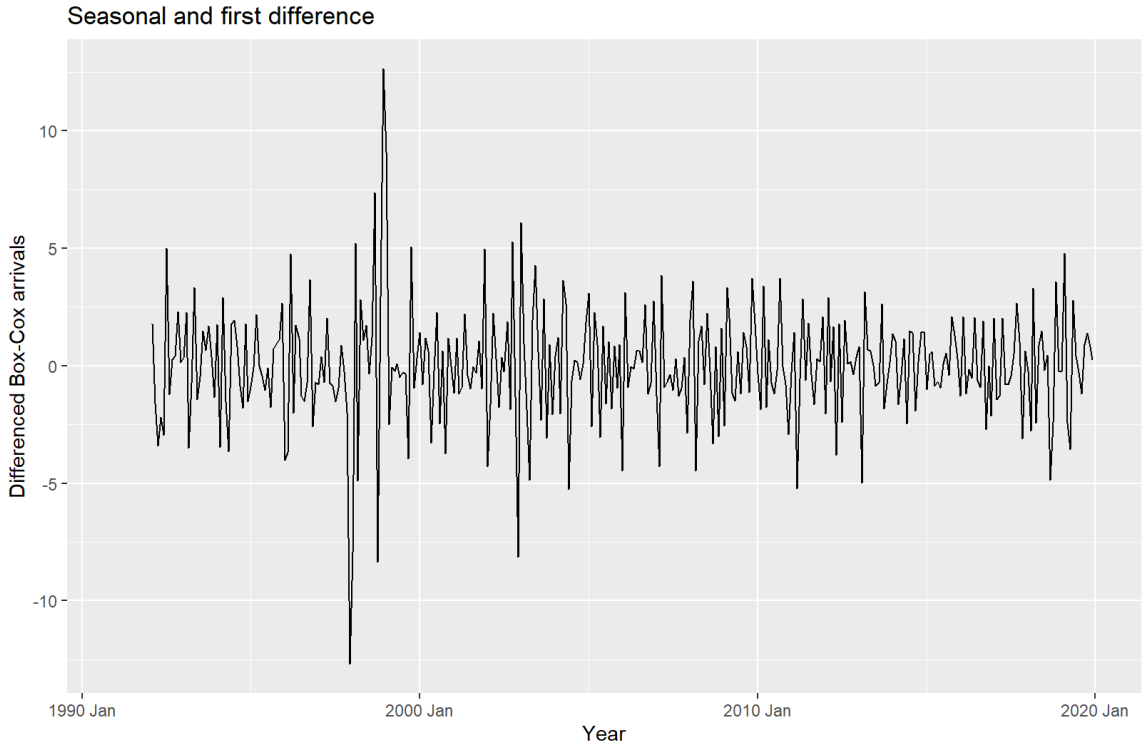


Figure 14. Series after both seasonal and first differencing (Step 4).

Applying both differences produces a series fluctuating around zero with no visible trend or seasonal pattern. The variance is roughly stable across the sample apart from the 1997 to 1998 crisis spike. The visual evidence supports the conclusion that d = 1 and D = 1 achieves stationarity.

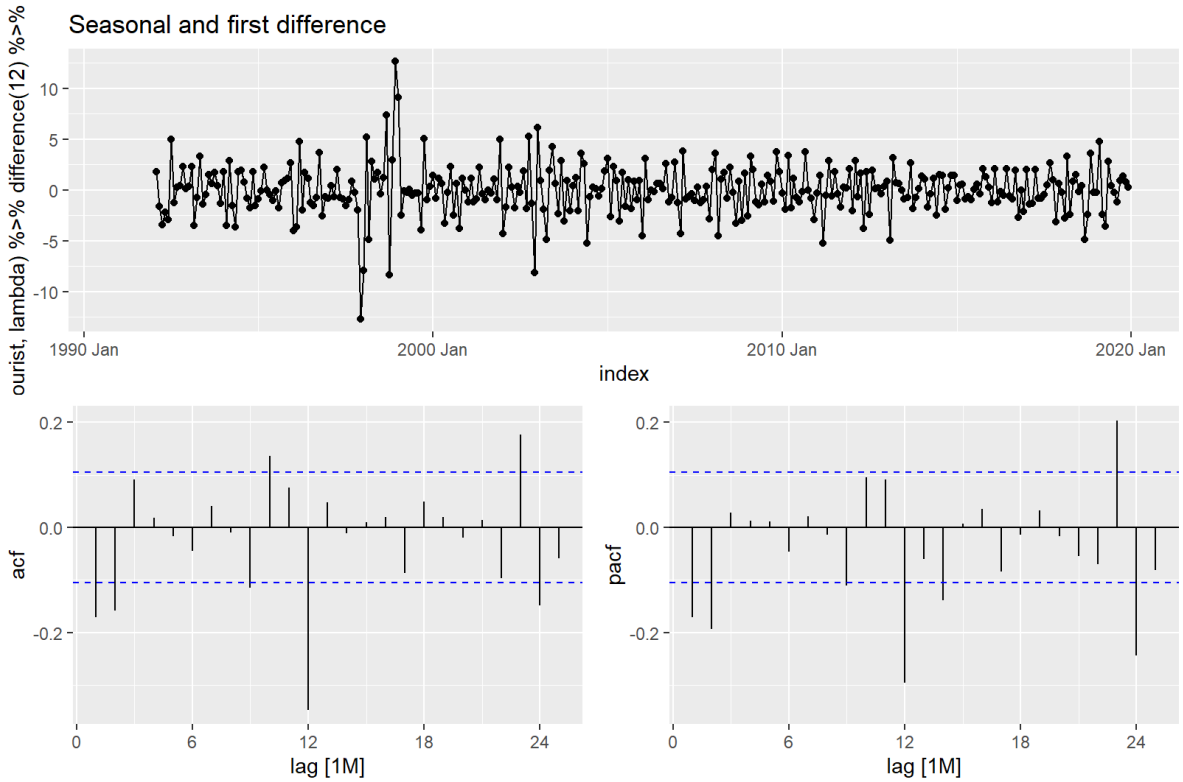


Figure 15. ACF and PACF after both seasonal and first differencing (Step 4).

The ACF shows a negative spike at lag 1 of approximately -0.20 followed by a rapid return toward zero, consistent with a non-seasonal MA(1) component. A moderate positive spike at seasonal lag 12 sits near the significance boundary, pointing to a seasonal MA(1) term. The PACF mirrors this with a negative spike at lag 1 and a spike at lag 12. These patterns suggest an ARIMA specification with non-seasonal MA terms and a seasonal MA(1). Note that the differencing analysis here uses d = 1 and D = 1 based on the unit-root tests. The auto-ARIMA in Question 6b applies its own internal criterion to the seasonally differenced series and selects d = 0 and D = 1. Both specifications are compared via AICc in Question 6b, where the d = 0 model proves superior.

Question 6b

Auto-ARIMA selected ARIMA(1,0,3)(1,1,1)[12] on the Box-Cox transformed series, with AIC = 1447.26, AICc = 1447.60, and BIC = 1473.98. The estimated coefficients are ar1 = 0.9701 (se = 0.0191), ma1 = -0.1875, ma2 = -0.1428, ma3 = 0.0721, sar1 = 0.2215, and sma1 = -0.8231, with sigma squared = 4.117.

The model uses d = 0 rather than d = 1. Although the KPSS test on the undifferenced transformed series suggested ndiffs = 1, the auto-ARIMA algorithm applies its own unit-root test to the seasonally differenced series (D = 1) and finds that one seasonal difference is sufficient for stationarity. The near-unit-root AR(1) coefficient of 0.9701 captures any residual

level persistence without an additional first difference. This choice is vindicated by the AICc: every manual candidate based on d = 1 and D = 1 yields a higher AICc than the d = 0 specification, as shown in Table 1.

Table 1. Manual ARIMA candidates ranked by AICc (Box-Cox transformed series, D = 1).

Model	AICc
ARIMA(2,1,0)(0,1,1)[12]	1448
ARIMA(0,1,2)(0,1,1)[12]	1449
ARIMA(1,1,1)(0,1,1)[12]	1451
ARIMA(0,1,1)(0,1,1)[12]	1454
ARIMA(1,1,0)(0,1,1)[12]	1459
ARIMA(0,1,1)(1,1,0)[12]	1503

Every manual candidate has a higher AICc than the auto-ARIMA result of 1447.60. The confirmed degrees of freedom for the Ljung-Box test in Question 8 is p + q + P + Q = 1 + 3 + 1 + 1 = 6, matching the value used in the code.

Question 7

The ETS model is specified as ETS(A,A,A), fitted to the Box-Cox transformed series with lambda = 0.2688.

Additive errors. The Box-Cox transformation stabilises the variance so that error size is approximately constant. Additive errors are appropriate on this transformed scale.

Additive trend. The time plot (Figure 1) documents sustained upward growth throughout the sample. The trend smoothing parameter beta = 0.0001 is essentially zero, indicating the estimated slope is very stable and barely updated period to period, consistent with the smooth long-run trajectory in the data.

Additive seasonality. After the Box-Cox transformation, the seasonal amplitude is approximately constant across years. A proportional seasonal specification is not needed on the transformed scale. The seasonal smoothing parameter gamma = 0.1263 indicates moderate seasonal updating. The level smoothing parameter alpha = 0.7453 is high, meaning the level adapts quickly to incoming observations.

Question 8

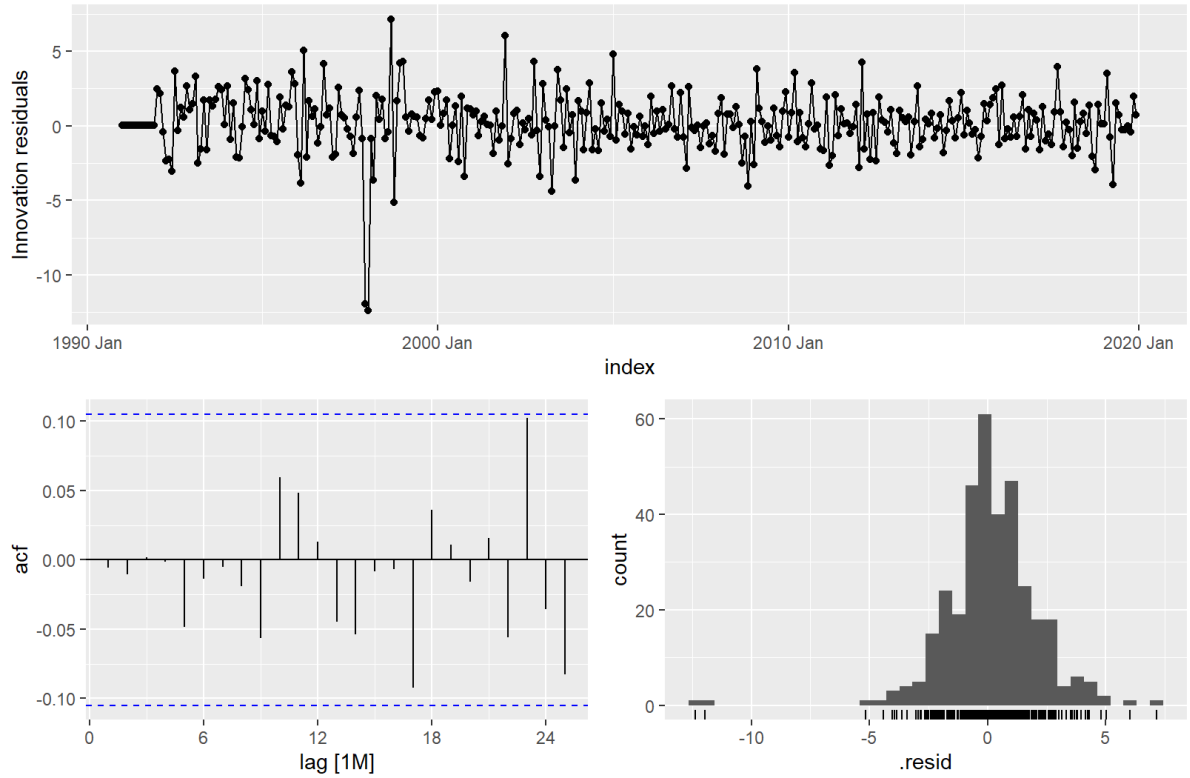


Figure 16. Residual diagnostics for the ARIMA(1,0,3)(1,1,1)[12] model.

The ARIMA residual time plot is centred close to zero throughout the sample. The single notable outlier is the 1997 to 1998 crisis residual at approximately -12 on the transformed scale. This is an isolated spike rather than part of any cluster, and the residuals before and after it show no systematic pattern. In the ACF panel, the bounds sit at approximately plus or minus 0.10, and no bar clearly exceeds them. The largest positive bar is at lag 12 at approximately 0.06, and the bars at lags 24 to 25 reach approximately -0.10, touching but not crossing the lower boundary. All other bars remain well within the bounds. The histogram concentrates between approximately -3 and +4 with a peak count near 60 at residuals close to zero. The distribution is approximately symmetric and broadly consistent with normality, with no heavy tails outside plus or minus 5 apart from the single crisis outlier. The Ljung-Box statistic is 4.43 with p-value = 0.618 (dof = 6, lag = 12; dof confirmed as 1+3+1+1 for this model). The white noise null is not rejected. The ARIMA model adequately captures the dependence structure of the series.

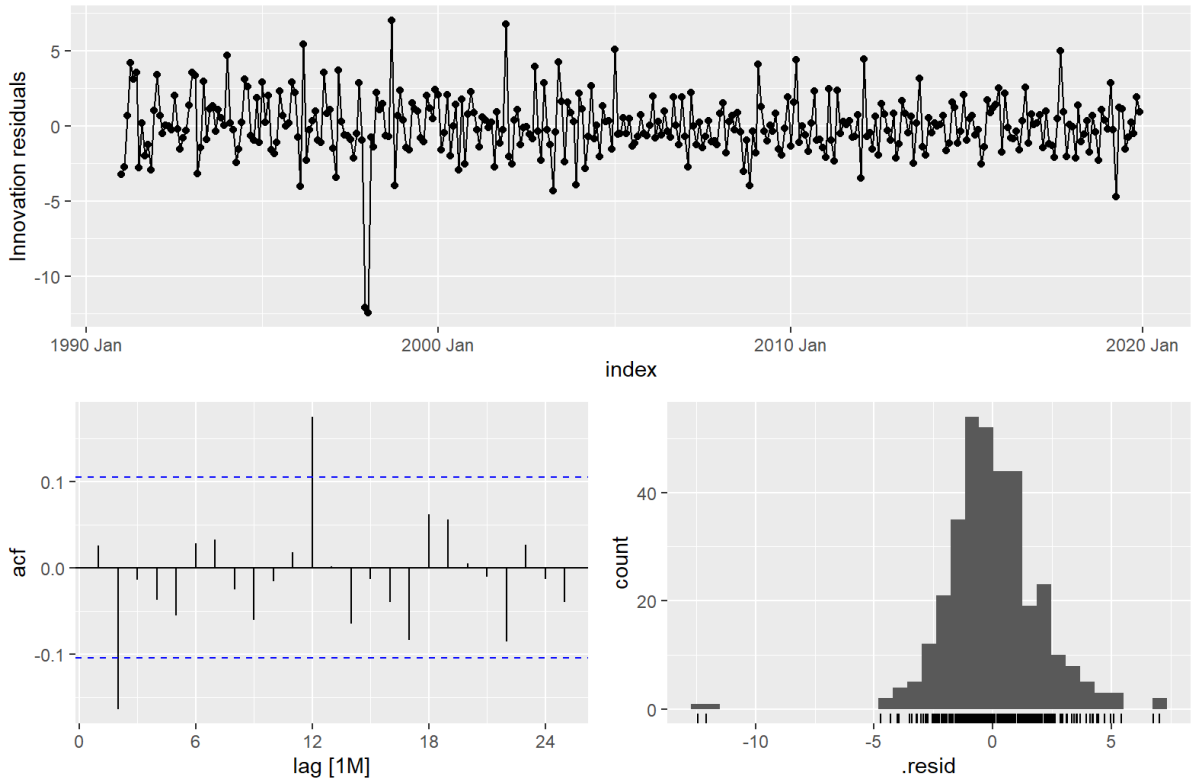


Figure 17. Residual diagnostics for the ETS(A,A,A) model.

The ETS residual time plot has a similar appearance to Figure 16, centred near zero with a crisis-period outlier at approximately -12 on the transformed scale. The difference becomes clear in the ACF panel. The bounds are again at approximately plus or minus 0.10, but the bar at lag 12 reaches approximately 0.15, exceeding the upper significance bound by around 50%. Bars at lags 17 and 18 are elevated at approximately 0.06 to 0.07, close to the boundary but within it. Short-lag autocorrelations (lags 1 to 6) are small and mostly negative, reflecting an over-differencing effect rather than systematic short-range dependence, but the clearly significant lag-12 spike is enough to reject white noise. The histogram resembles that of the ARIMA model in its central shape but the left tail extends to approximately -12 from the crisis outlier. The Ljung-Box statistic is 24.9 with p-value = 0.000358 (dof = 6, lag = 12), falling well below the 1% threshold. The white noise null is rejected. The ETS model fails to capture the seasonal autocorrelation at lag 12, meaning its prediction intervals incorporate uncertainty that partly reflects systematic residual dependence rather than genuine future randomness. On residual adequacy, ARIMA is the superior model.

Question 9

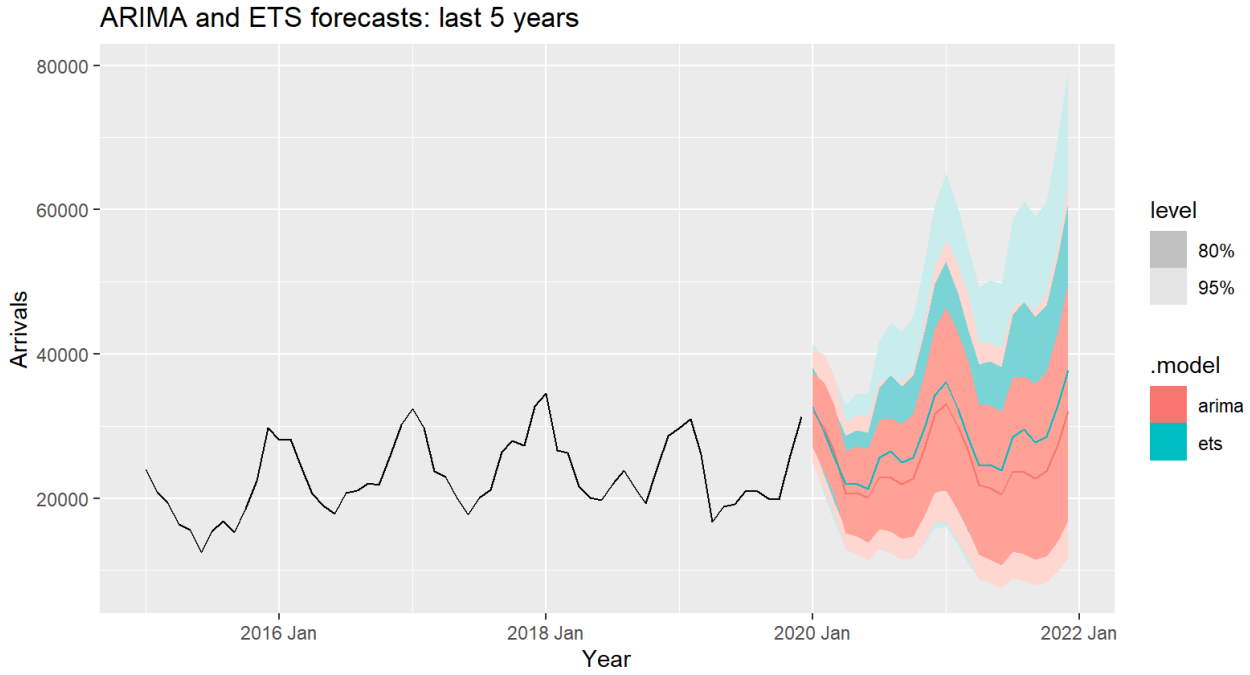


Figure 18. Two-year forecasts from ARIMA(1,0,3)(1,1,1)[12] and ETS(A,A,A), last five years of history shown.

Both models project the W-shaped seasonal cycle forward from January 2020, with peaks in winter months and troughs in mid-year. In the first forecast month (January 2020), the ARIMA point forecast is 32,114 arrivals and both models are closely aligned. The gap between them widens over time: by the second-year winter peak in December 2021, the ARIMA point forecast is approximately 35,000 arrivals while the ETS forecast is roughly 5,000 arrivals higher. The most striking difference is in the prediction intervals. By December 2021, the ARIMA 95% upper bound reaches approximately 52,000 arrivals while the ETS 95% upper bound extends to approximately 75,000 arrivals, a gap of around 23,000. The ARIMA lower bound dips close to zero in the mid-year trough months of 2020, while ETS lower bounds remain somewhat higher. This contrast is consistent with the ARIMA model having a lower residual variance (sigma squared = 4.12 versus 4.33 for ETS) and passing the Ljung-Box test where ETS does not. The tighter and better-calibrated ARIMA intervals are a direct consequence of its superior residual adequacy.

PART C - MODEL EVALUATION

Question 10

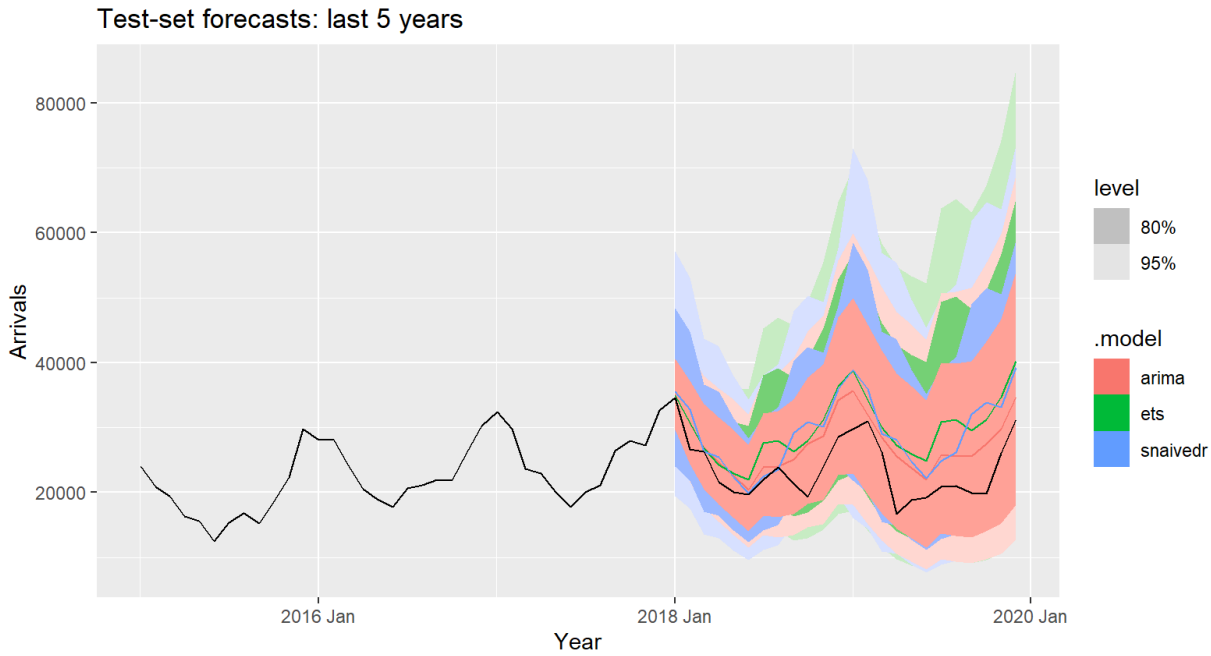


Figure 19. Test-set forecasts from the seasonal naive, ARIMA and ETS models against observed arrivals, last five years shown.

The training set covers January 1991 to December 2017 (348 months) and the test set covers January 2018 to December 2019 (24 months). Figure 19 plots all three model forecasts against the actual observed arrivals during the test period, shown as the black line.

ARIMA (red) tracks the actual values most closely. Its point forecasts remain nearest to the observed monthly counts throughout and its prediction intervals are the narrowest of the three. The seasonal naive with drift (blue) and ETS (green) both over-forecast, with their point forecasts sitting clearly above the actual data, particularly around the seasonal peaks of late 2018 and late 2019. The ETS fan is the widest overall, consistent with its larger residual variance and the unaddressed lag-12 autocorrelation identified in Question 8.

All three models share a systematic positive bias: their point forecasts exceed the actual values for most of the test period. The models were trained on three decades of consistent growth and naturally projected continued growth into 2018 to 2019. What actually occurred was a plateau and modest decline in South Korean arrivals to Australia. Rising competition from Japan as a short-haul Korean travel destination, a relatively strong Australian dollar, and a maturing outbound travel market all contributed to softer demand than the historical trend implied. ARIMA overshoots the least.

Question 11

Table 2. Test-set forecast accuracy metrics for the three models evaluated against the 2018 to 2019 holdout.

Model	MASE	RMSE	MAE
ARIMA	1.21	4,491	3,813
Seasonal Naive with Drift	1.78	6,868	5,607
ETS	1.97	7,013	6,227

ARIMA achieves the lowest MASE (1.21), RMSE (4,491) and MAE (3,813) across all three accuracy measures, making it the best-performing model on the test set. MASE is the primary comparison criterion because it is scale-independent: it expresses each model's mean absolute forecast error relative to the in-sample average error of a seasonal naive benchmark. Values below 1 indicate outperforming that benchmark; values above 1 indicate falling short.

All three models record MASE above 1, and all three produce negative mean errors (ARIMA: -3,813; seasonal naive: -5,587; ETS: -6,227), confirming systematic over-prediction throughout 2018 to 2019. As discussed in Question 10, this reflects the structural deceleration in South Korean arrivals that no model trained on pre-2018 growth data could anticipate. Despite this limitation, the ranking is clear: ARIMA's forecast error is 32% lower than the seasonal naive (MASE 1.21 versus 1.78) and 39% lower than ETS (1.21 versus 1.97). ARIMA is the recommended model for short-term forecasting of South Korean visitor arrivals to Australia.